Carter Montag

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Research Interests

Mathematical physics; resonance and scattering theory; nonadiabatic ionization; strong-field laser–matter interactions; atomic, molecular, and optical physics; scientific computing.

Education

University of Arizona *Aug 2024 – Present*

PhD in Applied Mathematics, expected May 2028

MS in Optical Sciences, expected May 2027

- **Advisor:** Miroslav Kolesik
- **GPA:** 3.93/4.00
- **Selected coursework:** Algorithms and High Performance Computing; Hilbert and Banach Spaces; Perturbative Methods; Quantum Mechanics; Quantum Optics; Atom Optics; Computational Nonlinear Optics; Laser Physics; Computational Optical Physics

Indiana State University *May 2022 – May 2024*

MS in Mathematics, GPA 3.97/4.00

Colorado State University *Aug 2012 – May 2015*

BS in Physical Science; minors in Mathematics and Chemistry

Research Experience

Doctoral Researcher *Tucson, AZ*

University of Arizona *Aug 2024 – Present*

- Investigate nonadiabatic ionization in long-wavelength laser fields, with the goal of developing a mathematical foundation for the metastable electronic state approach (MESA) and its post-adiabatic corrections.
- Derived energy-normalized scattering states and characterized the spectral density and resonance structure of a one-dimensional delta-potential model in an external electric field.
- Derived an evolution operator governing overlaps between instantaneous scattering states, providing a framework for describing nonadiabatic continuum coupling.
- Developed numerical methods to investigate continuum completeness, biorthogonality, and overlap propagation, validating analytical results and laying the groundwork for time-dependent simulations.

Laser Modeling Intern *Livermore, CA*

National Ignition Facility, Lawrence Livermore National Laboratory *May 2026 – Aug 2026*

- Developed and validated a neural-network surrogate mapping desired laser entrance hole (LEH) pulse shapes to master oscillator room (MOR) input waveforms for the National Ignition Facility.
- Achieved less than 0.6% mean RMS prediction error and several orders of magnitude faster evaluation than conventional physics simulations.
- Presented the project at the LLNL Summer Symposium and received the Judges' Choice Award.

Computational Physics Intern *Adelphi, MD*

Neutral Atom/Quantum Sciences Group, Army Research Laboratory *May 2025 – Aug 2025*

- Designed and implemented a hybrid analytic–numerical Doppler-averaging solver for the open-source RydIQule package used to model Rydberg-atom RF sensors.
- Achieved a 36× speedup and 64× lower memory usage for 2D simulations; enabled previously infeasible 3D simulations with a 61× lower memory footprint.

Evidence-Theoretic Periodicity Detection *Tucson, AZ*

Raytheon-Sponsored Research Project, University of Arizona *Spring 2026*

- Developed a time-series analysis pipeline combining persistent homology, spectral methods, autocorrelation, wavelet, and Lomb–Scargle diagnostics through Dempster–Shafer evidence fusion.
- Constructed a benchmark of 6,000 periodic and aperiodic signals with simulated sensor dropout and identified autocorrelation as the strongest baseline, with 97.15% in-sample accuracy.
- Implemented an OpenCL-based evidence-fusion workflow using fast transforms over mass-function representations to support larger evidence frames and repeated fusion workloads.

Simulation Relations for Finite Automata *Terre Haute, IN*

Graduate Research Project, Indiana State University *Fall 2023*

- Studied simulation relations between nondeterministic finite automata using a structural formulation of their state-transition systems.
- Proved that the existence of a simulation from one automaton to another implies inclusion of the language accepted by the first automaton in the language accepted by the second.

Publications

B. N. Miller, D. H. Meyer, **C. A. Montag**, O. Nagib, T. Virtanen, P. K. Elgee, and K. C. Cox, "RydIQule Version 2: Enhancing graph-based modeling of Rydberg atoms," *Journal of Open Source Software*, vol. 11, no. 118, p. 8539, 2026. doi:10.21105/joss.08539

Presentations

- **C. Montag**, J. Murphy, "Accelerated Data Fusion Algorithms," presented at the Annual Raytheon Day Symposium, Tucson, AZ, 2026.
- **C. Montag**, S. McLaren, and M. Erickson, "Machine-Learned Inverse Pulse Solving for NIF Pulse Shape Prediction," poster presented at the Lawrence Livermore National Laboratory Summer Symposium, Livermore, CA, 2026. Judges' Choice Award.
- **C. Montag**, "Modeling Rydberg Atomic Sensors," poster presented at the Department of Defense Institutes of Higher Education Opportunities Workshop, Washington, DC, 2025.

Awards and Fellowships

- GEM Employer PhD Fellowship, sponsored by Lawrence Livermore National Laboratory 2026–2027
- Judges' Choice Award, Lawrence Livermore National Laboratory Summer Symposium 2026

Teaching Experience

Mathematics Instructor *Remote*
Art of Problem Solving *Sep 2024 – Present*
- Teach advanced mathematics through inquiry-based online courses, guiding students to formulate, compare, and communicate independent solution strategies.

Graduate Teaching Assistant *Tucson, AZ*
University of Arizona *Aug 2024 – Dec 2025*
- Led review sessions and office hours for college algebra and business calculus, graded assignments, and provided targeted feedback in coordination with the lead instructor.

Mathematics Teacher and Department Chair *Denver, CO*
Bishop Machebeuf High School *Aug 2021 – Jun 2024*
- Led departmental functions, designed mathematics curricula, and served on academic and ethics committees addressing student performance and academic integrity.
- Taught mathematics to a diverse student population and coordinated communication among faculty, students, and families.

Music Teacher, *Our Lady of Lourdes Classical School* *Denver, CO*
Aug 2018 – Jun 2020

Chemistry Teaching Fellow, *BASIS Charter School* *Phoenix, AZ*
Aug 2016 – Jun 2017

Additional Professional Experience

Assistant Music Director *Englewood, CO*
Holy Name Church *Jul 2021 – Jul 2024*
Quality Control Chemist *Fort Collins, CO*
TOLMAR, Inc. *Jun 2015 – Jul 2016*

Technical Skills

Programming and tools: Python, MATLAB, PyTorch, OpenCL, Git, LaTeX, C++.
Methods: resonance and scattering theory, asymptotic analysis, time-dependent quantum simulation, numerical linear algebra, scientific machine learning, high-performance computing, persistent homology, Dempster–Shafer evidence fusion.